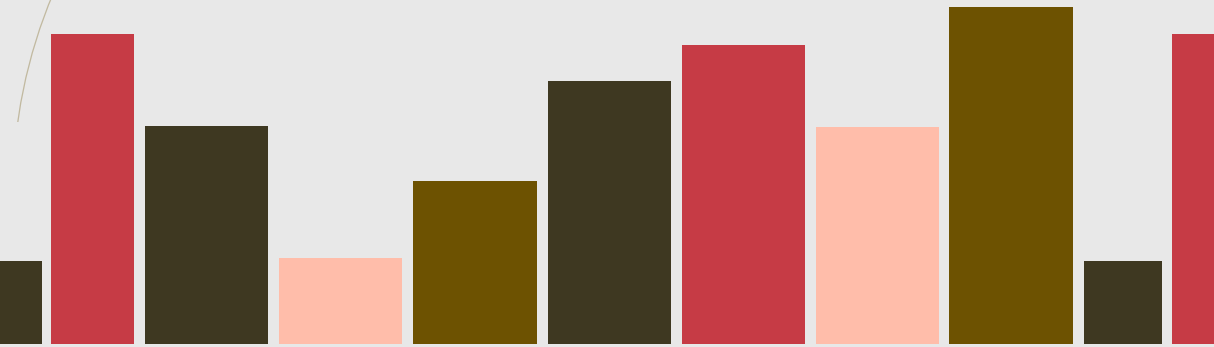
Unit #2



February 4th

Dense Encoding

=> use 1 qubit to send info of 2+ qubits (if entangled)

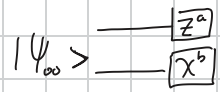
General Bell States:

$a, b \in \{0, 1\} \rightarrow 4$ possibilities 00, 01, 10, 11

$$|\psi_{ab}\rangle = \hat{Z}^a \otimes \hat{X}^b \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$= Z_1^a X_2^b \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = Z_1^a X_2^b |\psi_{00}\rangle$$

Circuit diagram:

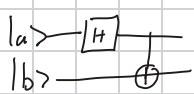


If we have:

a	b	$Z^a \otimes X^b$
0	0	$Z^0 \otimes X^0 = I \otimes I \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$
0	1	$I \otimes X \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \frac{1}{\sqrt{2}}(\|10\rangle + \|01\rangle)$
1	0	$Z \otimes I \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$
1	1	$Z \otimes X \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$

4 bell states

General Bell States:



ex $|00\rangle$

$$|00\rangle \xrightarrow{H \otimes I} |+0\rangle \xrightarrow{CNOT_{1,2}} \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

ex $|01\rangle$

$$|01\rangle \xrightarrow{H \otimes I} |+1\rangle \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

Use equivalence to have super close coding

Clicker:

Clicker

If Alice sends Bob a single classical bit, how many possible messages can she transmit?

A. 0
B. 1
C. 2 → 0 → false, 1 → true
D. 4
E. something else

Quantum Dense encoding:

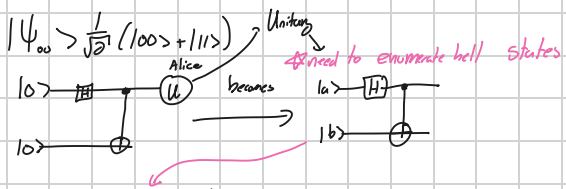
each have 1 qubit from $|\psi_{00}\rangle$

Alice & Bob share an entangled state: $|\psi_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Alice = A
Bob = B

→ Alice wants to send 2 bits to Bob
→ A operates only on her qubit
→ A sends her qubit to B
→ B measures these 2 qubits
→ using unitary operations to measure (not directly)

Distribute Bell State



$$|\psi_{ab}\rangle = Z^a \otimes X^b |\psi_{00}\rangle$$

Property of Bell States

Need to show: $I \otimes X |\psi_{00}\rangle = X \otimes I |\psi_{00}\rangle$

→ $I \otimes X \neq X \otimes I$ ... but also we have $|\psi_{00}\rangle$ which changes things (our bell state :))

LHS $I \otimes X \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

RHS: $X \otimes I \ \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$ } equivalent

Generalization: (will not cover in depth)

$U \otimes I \ |\psi_{00}\rangle = \pm\, \mathbb{I} \otimes U^T |\psi_{00}\rangle$

Procedure

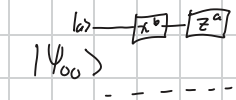
Now Alice opperate on her qubit

$|\psi_{ab}\rangle = Z^a \otimes X^b |\psi_{00}\rangle$

$= (Z^a \otimes I)(I \otimes X^b)|\psi_{00}\rangle$

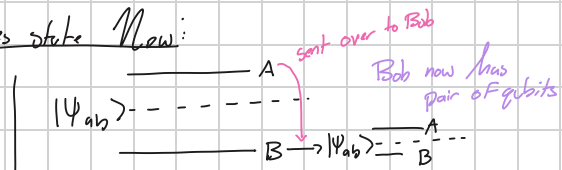
$= (Z^a \otimes I)(X^b \otimes I)|\psi_{00}\rangle$

Creat



If alice wants to send the info she can use X opperator

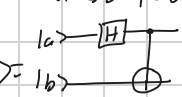
Alices state Now:



Bob now has pair of Fqubits

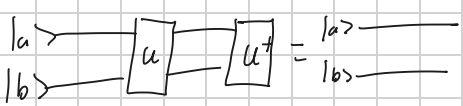
→ Now how does Bob get the info

Bob $|\psi_{ab}\rangle$ =



* Both CNOT & Hadamard are unitary

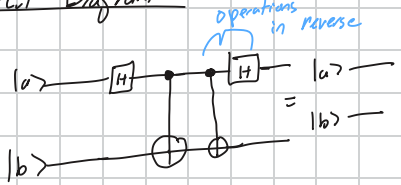
Bob can apply U^+



Identity $= (AB)^+ = B^+ A^+$

Let $\boxed{A}$ be A & $B = CNOT_{1,2}$

Circuit Diagram



Measurements??

Suppose Alice & Bob have $|\Psi_{ab}\rangle$

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

We know that:

↳ partial measurement

↳ 2nd bit $P(0) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$ $P(1) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$

$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ 2nd bit $P(0) = \frac{1}{2}$ $P(1) = \frac{1}{2}$

$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ $P(0) = \frac{1}{2}$ $P(1) = \frac{1}{2}$

Bell states have maximum entanglement, so if only one person measures, then the result is totally uncorrelated

→ Need to measure both A to be useful!

Review + Clarifying Information:

A & B $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Psi_{00}\rangle$

→ Alice applying a state to Bob = Bob altering his own state

Alice applies $Z^a X^b \otimes I$ on $|\Psi_{00}\rangle$

If Alice doesn't send qubit → there is some superluminal/random? transmission

If Bob has $|\Psi_{ab}\rangle$, b can decode

Bob can also send a qubit to Alice (since state is symmetric!)

2/6 Lecture Notes

Quantum Dense Encoding

A & B share $|\Psi_{00}\rangle = |0\rangle$ — H — A, $|0\rangle$ — B

A applies $Z^a X^b$ (2 bits) on 1st qubit $|\Psi_{00}\rangle$

$|\Psi_{ab}\rangle = (Z^a X^b \otimes I)|\Psi_{00}\rangle =$ $|a\rangle$ — H, $|b\rangle$

A sends her qubit to B, 1 qubit

B has the full state $|\Psi_{ab}\rangle$

B applies a decoding unitary

$|a\rangle$ — H — $|\Psi_{ab}\rangle$ — H = $|a\rangle$

$|b\rangle$ — $|b\rangle$

Quantum Teleportation

A + B get together, share $|\Psi_{00}\rangle$

Carol wants A to send C qubit to B

After this, no quantum communication is allowed

A + C together do some measurement and send result to B ∈ two qubits
↓
classical

B will apply unitary to recover C's info

Q.D.E: sending 2 bits classical by transmitting 1 qubit

Q.T: sending 1 qubit information by transmitting 2 classical bits

↳ BOTH USE $|\Psi_{00}\rangle$

- A + B share a Bell state:

$$|\Psi_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle_a|0\rangle_b + |1\rangle_a|1\rangle_b$$

$$\neq |\Psi_a\rangle|\Psi_b\rangle \text{ (NOT seperable)}$$

- C's qubit

$$|\Psi\rangle_C = \alpha|0\rangle_C + \beta|1\rangle_C$$

- Total State is $|\Psi\rangle_C \otimes |\Psi_{00}\rangle$ (3 qubits)

$$|\Phi\rangle_{abc} = (\alpha|0\rangle_c + \beta|1\rangle_c) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{ab} + |11\rangle_{ab})$$

Note $\hbar$ is measurement

① $CNOT_{C,A}\,|\Phi\rangle_{CAB} = \alpha|0\rangle_C \cdot \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$

$$+ \beta|1\rangle_C \cdot \frac{1}{\sqrt{2}}(|10\rangle_{AB} + |01\rangle_{AB})$$

② $H_C = H \otimes I \otimes I$

$$\alpha \frac{1}{\sqrt{2}}(|0\rangle_c + |1\rangle_c)\frac{1}{\sqrt{2}}(|00\rangle_{ab} + |11\rangle_{ab})$$

$$+ \beta \frac{1}{\sqrt{2}}(|0\rangle_c + |1\rangle_c)\frac{1}{\sqrt{2}}(|10\rangle_{ab} + |01\rangle_{ab})$$

$$= \frac{1}{2}|00\rangle_{ca} \otimes (\alpha|0\rangle_b + \beta|1\rangle_b)$$

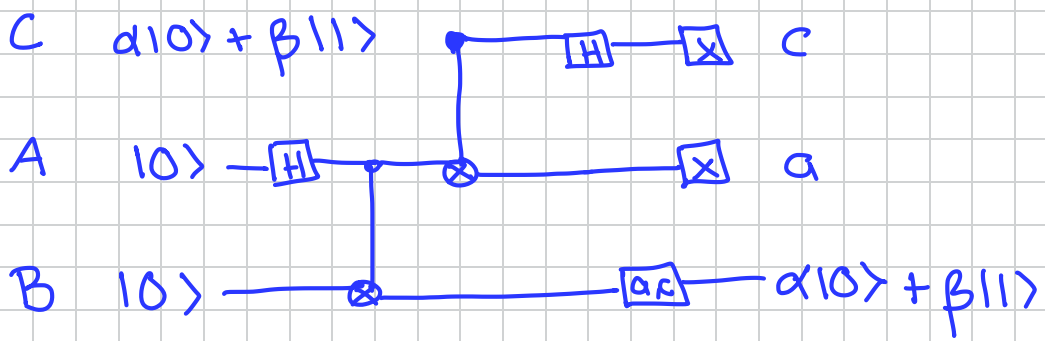
$$+ \frac{1}{2}|01\rangle_{ca} \otimes (\alpha|1\rangle_b + \beta|0\rangle_b)$$

$$+ \frac{1}{2}|10\rangle_{ca} \otimes (\alpha|0\rangle_b - \beta|1\rangle_b)$$

$$+ \frac{1}{2}|11\rangle_{ca} \otimes (\alpha|1\rangle_b - \beta|0\rangle_b)$$

Measure C & A		B's state
C	A	
0	0	$\alpha\|0\rangle_b + \beta\|1\rangle_b$
0	1	$\alpha\|1\rangle_b + \beta\|0\rangle_b$
1	0	$\alpha\|0\rangle_b - \beta\|1\rangle_b$
1	1	$\alpha\|1\rangle_b - \beta\|0\rangle_b$

Summary



C	a	B's state	B's correction/recovery
0	0	$\alpha\|0\rangle + \beta\|1\rangle$	I
0	1	$\alpha\|1\rangle + \beta\|0\rangle$	X
1	0	$\alpha\|0\rangle - \beta\|1\rangle$	Z
1	1	$\alpha\|1\rangle - \beta\|0\rangle$	$-XZ = ZX$

Quantum Cryptography

Introducing Alice & Bob:

⇒ want to send encrypted message

⇒ messages represented by binary

Question!
What is the standard for strings represented in binary for Quantum Computing?

Way we can encode this:

Suppose our input is	another bit	output
0	0	$0 \oplus 0 = 0$
0	1	$0 \oplus 1 = 1$
1	0	$1 \oplus 1 = 1$
1	1	$1 \otimes 1 = 0$
x	R → R value Matters	

output appear in public channel

⇒ can hide original information well

Can't brute force attack
⇒ cannot find out x or r easily
1/2 chance for each
→ having 1/2 probability for each is most secure

What if you XOR more than once

$r \oplus r$

$0 \oplus 0 = 0$

$1 \oplus 1 = 0$

$y = x \oplus (r \oplus r)$

$= x \oplus 0$

$= x$

$x \oplus r \oplus r = x$

Bob can apply $\oplus r$ to get initial information

But there is a problem

⇒ if you use same r, not secure
→ letter frequencies can be found, can deduce mapping

Alice + Bob can prochare but what if they never meet

→ sharing r securely over & over again is same as sharing x directly

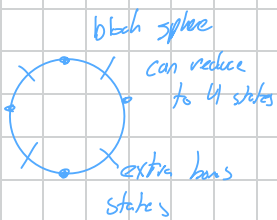
- A & B can share random r quantumly and detect any eavesdropping
  → what about No cloning?? How is r shared?
  → BB84 protocol (Bennet, Brassward)

→ measurement of quantum state collapses state
→ can detect tampering
→ Eve cannot copy info b/c no cloning

One time pad: How to get one

- Make two copies of a bunch of random strings.
- Use a trusted courier to send the pad to the person you want to talk to
- Put it in a locked briefcase on the way.
- Use a trusted courier to send the key.
- But. What if the courier is a bad guy.

Not great for security
1) need lots of different pads
2) can leave messages



The protocol:

- A sends B a long sequence of qubits randomly chosen in/of the 4 states

$|0\rangle, |1\rangle, |+\rangle = H|0\rangle; |-\rangle = H|1\rangle$

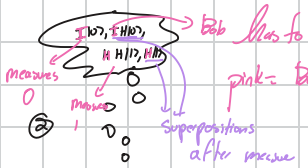
→ randomize to four choices minimum

- A sends to bit 0, and randomly sets $|0\rangle$ or $|+\rangle$
- A wants to send bit 1 random $|1\rangle, |-\rangle$
- B randomly each qubit through Measurement choose measurement bases randomly

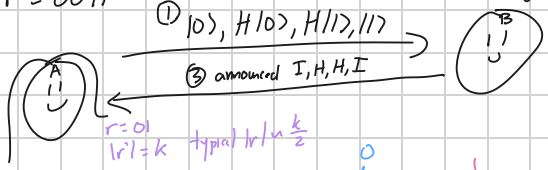
↓ changes probabilities to $\frac{1}{2}$ & $\frac{1}{2}$

12% flip better than 50% flip

The protocol



r = 0011

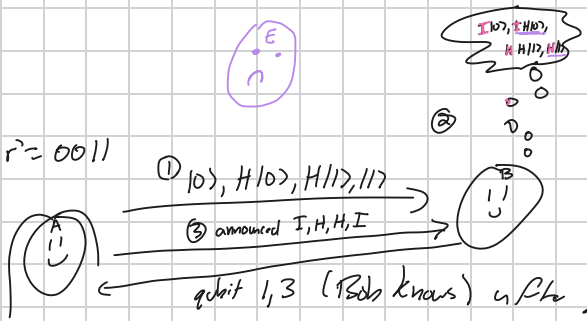


① Alice Randomly sends |0⟩, H|0⟩, |1⟩, H|1⟩

② Bob will randomly apply I or H on each qubit & Measured

③ Alice & Bob announce I or H

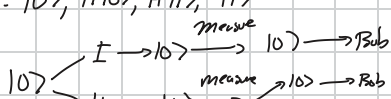
④ If I/H match then measurement result by bob is r



Assume Eve is honest but curious
→ too late for Eve to do anything here

Why Alice Chose H & I

ex: |0⟩, H|0⟩, H|1⟩, |1⟩



Eve has some probability of making things not match w/ some probability

Also, Eve might measure in the wrong basis

This probability is that Eve is detected

1/4 chance Eve's measurement gets detected

3/4 chance Eve's measurement undetected

for messages this probability is bound to happen at some point

Eve measures the qubits in some basis: I, H

$|0\rangle, |1\rangle$ } $\frac{1}{2}$ probability for each case
$H|0\rangle, H|1\rangle$ }

With $\frac{1}{4}$ probability to be detected

Alice can use a lot of such qubits to detect Eve.

Clicker:

Alice transmits $|1\rangle$
Eve Measures $\{|+\rangle, |-\rangle\}$
Bob Measures $\{|0\rangle, |1\rangle\}$
What is the probability Bob measures $|1\rangle$
c) 0.5

since $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$
$p = \frac{1}{2}$ $|+\rangle$
$p = \frac{1}{2}$ $|-\rangle$
if Bob measures $|+\rangle - p(1) = \frac{1}{2}$
$|-\rangle \rightarrow p(1) = \frac{1}{2}$

A perfect Attack
→ no cloning theorem important

Assume perfect copy Machine

Where $|0\rangle \rightarrow |00\rangle$
$|1\rangle \rightarrow |11\rangle$
$|+\rangle \rightarrow |++\rangle$
$|-\rangle \rightarrow |--\rangle$

If Eve can copy ... then this protocol is not safe, since she disturbs only her own copy

Protected by No-cloning theorem

Clicker question

$|0\rangle|0\rangle \to |0\rangle$

$|1\rangle|0\rangle \to |1\rangle|1\rangle$

$|+\rangle|0\rangle \to |+\rangle|+\rangle$

$|-\rangle|0\rangle \to |-\rangle|-\rangle$

If would violate state normalization not be unitary & violate causality

Pf for causality violation

$|00\rangle + |11\rangle = |++\rangle + |--\rangle$

Alice measures $|0\rangle|1\rangle \to$ ½ 0, ½ 1

Bob $|0\rangle|1\rangle$

Alice measures $|+\rangle|-\rangle$ ½ 0, and ½ 1

$|+\rangle|-\rangle$ ½ 0 & ½ 1

} continuum of one copy

If Bob can copy

$|0\rangle \to |0000000\rangle$

$|1\rangle \to |1111111\rangle$

$|+\rangle \to |+++++++\rangle$ → for the 3rd case Bob can get random 0's and 1's

If Bob copys qubits & does measurement

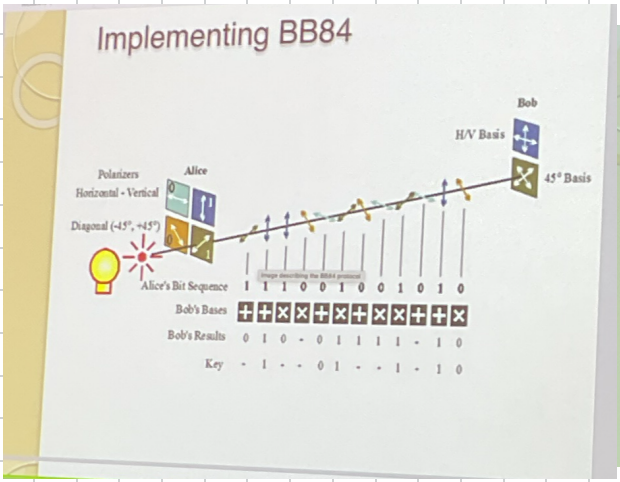
→ distribution changes

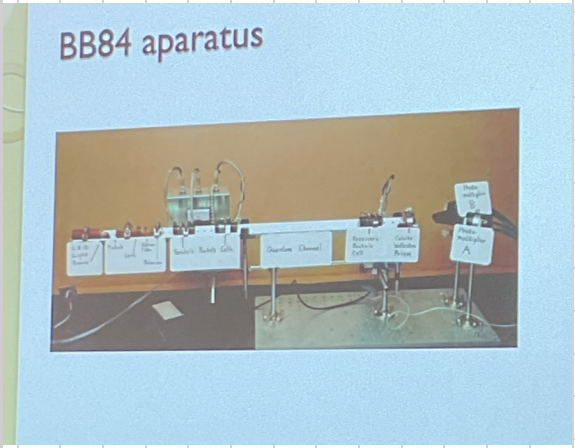
Alice can distinguish case if it doesn't violate causality

Summary:

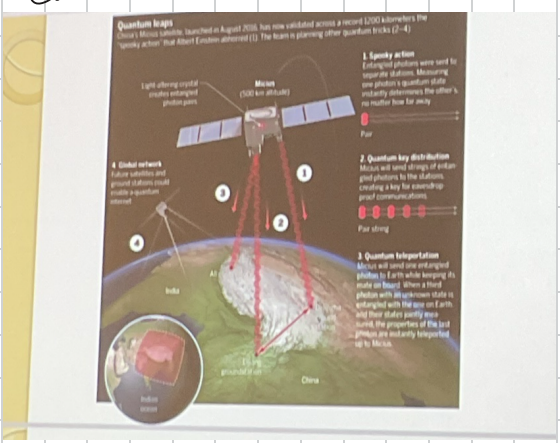
1) Alice sends $|0\rangle, |+\rangle, |1\rangle, |-\rangle$ randomly
2) Bob randomly applies I & H
3) A & B announce I/H, check whether basis matches for I/H
4) choose some of the qubits if I/A match
5) Measure & check 0/1 match
6) If every bit matches, the rest is r

Using photons (not really relevant)





Quantum Satellites :)



General computational process:

→ given a bitstring x, after some procedure get a new bitstring $f(x)$

$f : \{0,1\}^n \longrightarrow \{0,1\}^n$

corresponds from $x \longrightarrow f(x)$

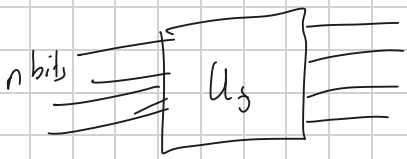
→ classical computation is a subset of quantum computation

Each of 2^n, 2^m bitstrings corresponds to a computational basis state.

$x = 0110$ $\quad |x\rangle = |0110\rangle$

f unitary (reversible computation)

Clicker: Is this circuit reversible reversible = 1 to 1



Not one to one

If $n > m$ (cannot be 1 to 1) even if $n = m$

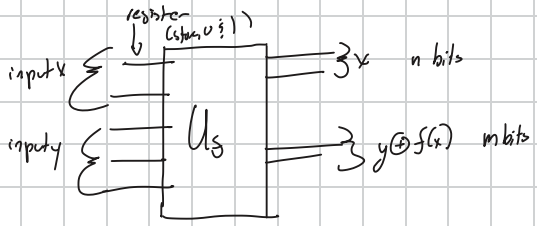
Suppose $f(x) = 0$

Reversible Architecture:

→ keep norm as 1 (P needs to be 1, and info cannot be lost)
→ info recovery: information is preserved
→ to enforce reversibility: writing will restrict to reversible
  → restrict to classical case
  → bc not all classical case is reversible

Kinda bad in explanation lol

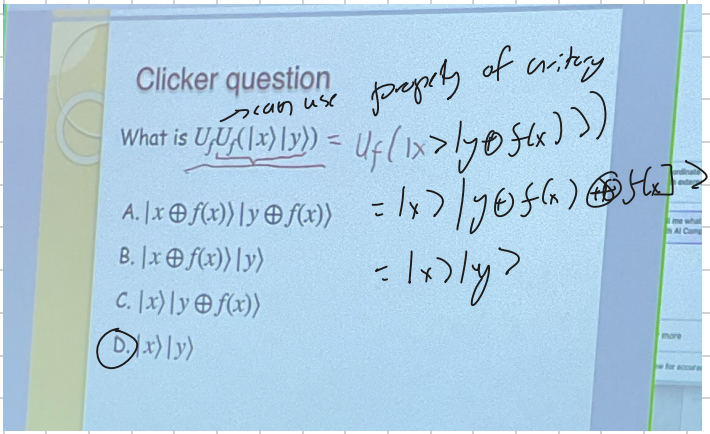
To enforce this:



$U_f(|x\rangle_n |y\rangle_m)$
$= |x\rangle_n |y \oplus f(x)\rangle_m$
n+m bits → n to n bits
→ $|x\rangle|z\rangle$

Suppose we see $z = y \oplus f(x)$

We can see $x \Rightarrow f(x)$ can compute $z \oplus f(x) = y \oplus f(x) \oplus f(x) = y$



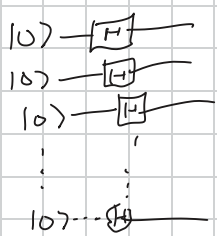
U_f must be reversible !!

n-fold Hadamard: Basis Superposition

consider $H^{\otimes n}|0\rangle_n$

$|0\rangle|0\rangle\cdots|0\rangle$ (n times)

$(H \otimes H \otimes \cdots \otimes H)$ (n times)



$(H|0\rangle)^{\times n}$ $n=1$

$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$n=2$

$\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$

$= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

Suppose $H^{\times n}|0\rangle_n = \frac{1}{\sqrt{2^n}} \sum_{0 \le x < 2^n} |x\rangle_n = \frac{1}{\sqrt{2^n}} \sum_{x_1, x_2, \ldots x_n \in \{0,1\}} |x_1, x_2, \ldots x_n\rangle$

Quantum Parallelism:

$U_f\left(H^{\otimes n} \otimes I^{\otimes m}|0\rangle_n|0\rangle_m\right)$

$= \frac{1}{\sqrt{2^n}} \sum_{x_1, x_2 \cdots x_n \in [0,1]} |x_1 x_2 \cdots x_n\rangle$

Quantum Parallelism

$U_f \left(H^{\otimes n} \otimes I^{\otimes m} |0\rangle_n |0\rangle_m \right)$

$= U_f \left(\frac{1}{2^{\frac{n}{2}}} \sum_{0 \le x < 2^n} |x\rangle_n |0\rangle_m \right)$

$= \frac{1}{2^{\frac{n}{2}}} \sum_{0 \le x < 2^n} \left(U_f |x\rangle_n |0\rangle_m \right)$

$= \frac{1}{2^{\frac{n}{2}}} \sum_{0 \le x < 2^n} |x\rangle_n |f(x)\rangle_m$ (Quantum parallelism

Seems to contain info of 2^n evaluation of $f(x)$

2/16/26

Deutsch-Jozsa Algorithm

⇒ explains why Q.C better than classical
⇒ clarify quantum parallelism

Review: Partial Measurement

→ use Born Rule
⇒ might need to change basis

Suppose we have $|\psi\rangle_{12}$ measure 1st qubit in a basis: $|b_1\rangle, |b_2\rangle, |b_3\rangle, \ldots$ (orthonormal)

We want to know $|b_i\rangle_1$ remaining state (if we get i, it means we project the 1st qubit to be)

↳ $\dfrac{(\langle b_i|_1 \otimes \mathbb{I}_2)\cdot|\psi\rangle_{12}}{|\langle b_i|_1 \otimes \mathbb{I}_2 \cdot |\psi\rangle_{12}|}$ (for 1 qubit)

→ on 2nd bit
vector ⊗ operator
(partial inner product)
(should be a column vector)

What is the probability to get $p(i) = |(\langle b_i|_1 \otimes \mathbb{I}_2)\cdot|\psi\rangle_{12}|^2$ (Generalized Born rule)

Ex $|\psi\rangle = \dfrac{|00\rangle+|11\rangle}{\sqrt{2}} = \dfrac{|0\rangle_1|0\rangle_2+|1\rangle_1|1\rangle_2}{\sqrt{2}}$

⇒ measure in $|+\rangle_1$ & $|-\rangle_1$ basis

computing remaining state: (assume we get $|+\rangle_1$) → for 1st qubit
(up to a normalization... simplification)

ignoring normalization

$(\langle +|_1 \otimes \mathbb{I}_2)\cdot(|0\rangle_1|0\rangle_2 + |1\rangle_1|1\rangle_2)$ ⟹ remaining term

by linearity of all matrices → $= (\langle +|_1 \otimes \mathbb{I}_2)\cdot|0\rangle_1|0\rangle_2 + (\langle +|_1 \otimes \mathbb{I}_2)\cdot|1\rangle_1|1\rangle_2$

$= \langle +|0\rangle_1 \otimes \mathbb{I}_2|0\rangle_2 + \langle +|1\rangle_1 \otimes \mathbb{I}_2|1\rangle_2$

$= \frac{1}{\sqrt{2}}|0\rangle_2 + \frac{1}{\sqrt{2}}|1\rangle_2 = \frac{1}{\sqrt{2}}(|0\rangle_2+|1\rangle_2) = |+\rangle_2$

Remaining state $|+\rangle_1|+\rangle_2$ ⟹ tensoring them together

If we get a measurement result $|b_i\rangle$,

We define projector $p_i = |b_i\rangle\langle b_i| \otimes I_2$

remaining total state (both qubit $I \otimes 2$): $\frac{P_i|+\rangle_{12}}{|P_i|\psi_{1,2}\rangle}$ (for 2 qubits)

Note: calculate w/o norm until it has been simplified

prob $p(i) = \langle\psi|_{12} P_i \cdot P_i|\psi\rangle_{12} = \langle\psi|_{12} P_i |\psi\rangle_{12}$

- $P_i \cdot P_i = P_i$ by property of projectors
- P_i not necessarily observable ∵ P_i is hermetian: $P_i^\dagger = (|b_i\rangle\langle b_i|)^\dagger \otimes I^\dagger = |b_i\rangle\langle b_i| \otimes I$

$p(i)$: expectation value; $\langle\psi|_{12}$: remaining state

Continued Quantum Parallelism:

$f: x \to f(x)$

$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle \Rightarrow$ no info lost

Problem: can we get $f(x)$ for any x

$U_f \frac{1}{\sqrt{2^n}} \sum_x |x\rangle|0\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle|f(x)\rangle$ (linearity)

Applied by $H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle$

↳ highly entangled state

The catch: in order to get info, we need to measure

Suppose we measure x ($|x\rangle$) and get x^* (remaining state)

We get $|x^*\rangle|f(x^*)\rangle \Rightarrow$ probabilistic b/c any given x^*, $p(x^*) = \frac{1}{2^n}$

Suppose we have 1 qubit

$\begin{cases} f(0) = 0 \\ f(1) = 1 \end{cases}$ $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$

Now onto **Deutsch-Jozsa Algorithm**

Problem: suppose $f: \{0,1\}^n \longrightarrow \{0,1\}$ ⇒ boolean function maps n qubit to true/false

→ not consider general f restriction on f:

2 cases: ⇒ two special cases

constant: $f(x)=0 \ \forall x$ or $f(x)=1 \ \forall x$ ⇒ $f(x)=f(y)$ for any x and y

balanced: $n_0 = \{$ number of x s.t. $f(x)=0\}$ $\quad n_0 = n_1 \Rightarrow f(x) \neq f(y)$

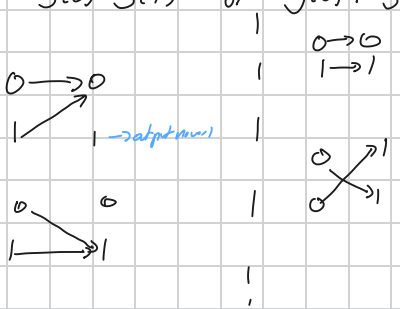
$n_1 = \{ \#x$, s.t. $f(x)=1\}$

Special cases of deutsch algorithm: ⇒ simplification

consider $f: \{0,1\} \longrightarrow \{0,1\}$ where $n=1$

const case or balanced? (guaranteed 1 of the cases)

$f(0)=f(1)$ or $f(0)\neq f(1)$

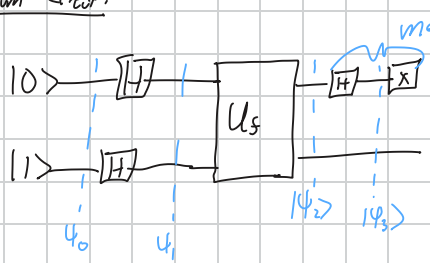


arrow = maps to

⇒ need to evaluate twice for this case classically

⇒ Q.M. only need once

Quantum Circuit:



$C=0 \Rightarrow f$ is const

$C=1 \Rightarrow f$ is const

① $|\psi_0\rangle = |0\rangle|1\rangle$

② $|\psi_1\rangle = |+\rangle|-\rangle$

What happens if we apply $U_f|\psi_1\rangle = U_f \frac{1}{\sqrt{2}} \sum_{x\in\{0,1\}^3} |x\rangle|-\rangle$

$$U_f(|x\rangle|-\rangle) = \frac{1}{\sqrt{2}}(U_f|x\rangle|0\rangle - U_f|x\rangle|1\rangle) \Rightarrow \text{by linearity}$$

$$= \frac{1}{\sqrt{2}}(|x\rangle|f(x)\rangle - |x\rangle|f(x)\oplus 1\rangle$$

$$= |x\rangle \frac{1}{\sqrt{2}}(|f(x)\rangle - |f(x)\oplus 1\rangle)$$

Applying $U_f|x\rangle|-\rangle$

If $f(x)=0$ $= |x\rangle\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle) = |x\rangle|-\rangle$

if $f(x)=1$ $= |x\rangle\frac{1}{\sqrt{2}}(|1\rangle-|0\rangle) = -|x\rangle|-\rangle$

$$U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$$

$$U_f(|x\rangle|0\rangle) = |x\rangle|f(x)\rangle$$

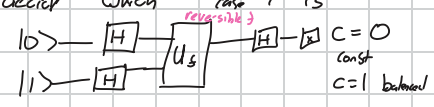
Special Case of Deutsch Algorithm

Function: $f: \{0,1\}^3 \to \{0,1\}^3$

First case:

$f(0) = f(1)$ constant or $f(0) \neq f(1)$ balanced case

$\Rightarrow$ decide which case f is



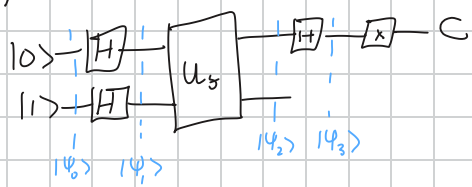
$$U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$$

want to utilize superposition

vs.

$$U_f(|x\rangle|0\rangle) = (|x\rangle|f(x)\rangle)$$

Analysis



$|\psi_0\rangle = |01\rangle$

$|\psi_1\rangle = H^{\otimes 2}|01\rangle = H|0\rangle \otimes H|1\rangle = |+\rangle|-\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|-\rangle)$

$|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2}}(U_f|0\rangle|-\rangle + U_f|1\rangle|-\rangle)$

$= \frac{1}{\sqrt{2}}(-1)^{f(0)}|0\rangle|-\rangle + (-1)^{f(1)}|1\rangle|-\rangle$

$= \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right)|-\rangle$ ⇒ not entangled

(the two coefficients have the same sign)

1st case $f(0) = f(1)$

$|\psi_2\rangle = (-1)^{f(0)} \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle$

$= (-1)^{f(0)}|+\rangle|-\rangle$

$= \pm|+\rangle|-\rangle$

2nd case $f(0) \neq f(1)$

$= \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right)|-\rangle$

(the two coefficients have different signs)

$= \frac{1}{\sqrt{2}}(-1)^{f(0)}(|0\rangle - |1\rangle)|-\rangle$

$= (-1)^{f(0)}|-\rangle|-\rangle = \pm|-\rangle|-\rangle$

Summary:

If ① $f(0) = f(1) \longrightarrow |\psi_2\rangle = \pm|+\rangle|-\rangle$
② $f(0) \neq f(1) \longrightarrow |\psi_2\rangle = \pm|-\rangle|-\rangle$

] orthonormal basis

Measure $|+\rangle|-\rangle$ for 1st qubit

$$|H\rangle = H \otimes I\, |\psi_2\rangle = \begin{cases} \pm(H|+\rangle)|-\rangle = \pm|0\rangle|-\rangle \\ \pm(H|-\rangle)|-\rangle = \pm|1\rangle|-\rangle \end{cases}$$

↓ by defn of Hadamard

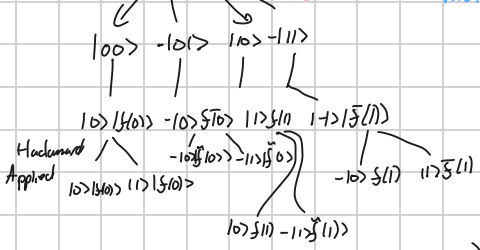
Why quantum Speed-up?

⇒ consider all computations / paths

Start from $|01\rangle$ state

Now we get $|01\rangle \longrightarrow |+\rangle|-\rangle$ (4 states)

Superposition means multiple paths at once

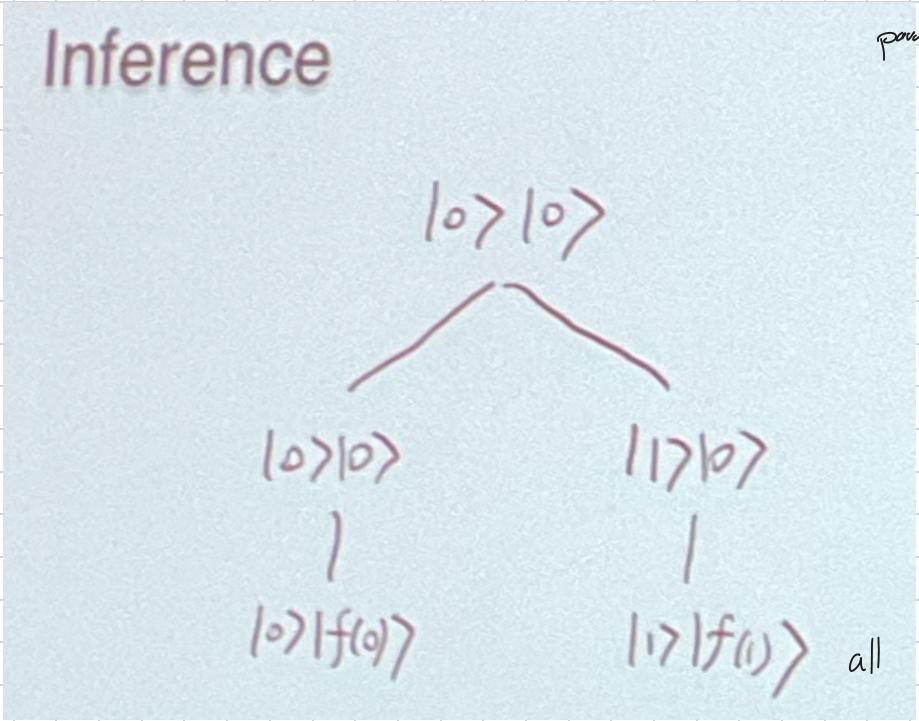


First-state $|0\rangle\left(|f(0)\rangle - |\bar{f}(0)\rangle\right) + |f(1)\rangle - |\bar{f}(1)\rangle\big)$

$|1\rangle\left(|f(0)\rangle - |\bar{f}(0)\rangle - |f(1)\rangle + |\bar{f}(1)\rangle\right)$

$f(0) \neq f(1)$ case $|1\rangle \cdot 0$

case $f(0) = f(1)$ $|1\rangle \cdot 0$



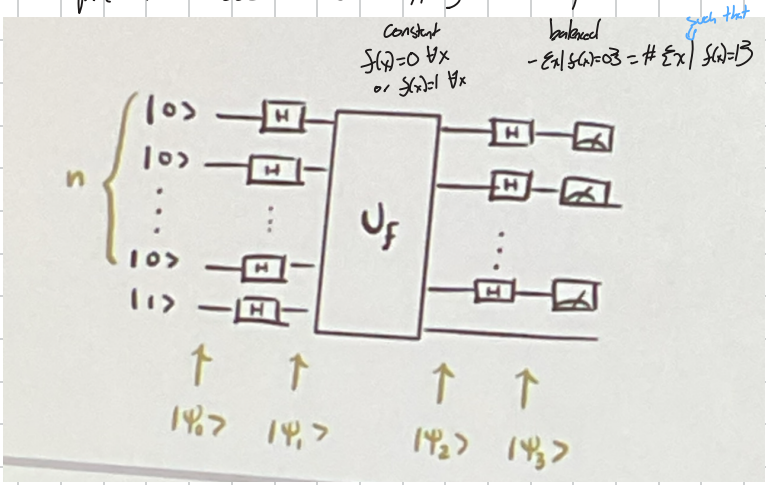
parallism not unique to quantum

no cancelation of different computational paths

cancelation due to quantum interference (Flavia double slit experiment)

SPEED-UP due to quantum interference

Complete Deutsch-Jozsa: $f:\{0,1\}^n \to \{0,1\}$ = Guaranteed two cases (?)



time Complexity

$O(2^{n-1}+1)$

suppose you checked 2^{n-1} have been checked

What the Fuck

=> check ½ and predomint

Deutsch–Jozsa Algorithm

$$f: \{0,1\}^n \longrightarrow \{0,1\}$$

constant $f(x)=0 \ \forall x$
$f(x)=1 \ \forall x$

balanced $\forall x$:
$\{\#x \mid f(x)=0\} = \#\{x \mid f(x)=1\}$

$$U_f |x\rangle |y\rangle = |x\rangle |f(x) \oplus y\rangle$$

Let us derive the intermediate steps:

$$\partial^{n-1}+1$$

Let us Analyze intermediate steps

$$|\psi_1\rangle = \frac{1}{\sqrt{\partial^n}} \sum_x |x\rangle |-\rangle$$

Note
$$(H|0\rangle)^{\otimes n} = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)^{\otimes n}$$

$$|\psi_2\rangle = U_f |\psi_1\rangle \qquad U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

extra phase

$$= \frac{1}{\sqrt{\partial^n}} \sum_x U_f |x\rangle |-\rangle = \frac{1}{\sqrt{\partial^n}} \sum_x (-1)^{f(x)} |x\rangle |-\rangle$$

$x \in \{0,1\}^n \Longrightarrow$ two representations $\rightarrow$ bit string $x = x_1, x_2 \dots x_n$

$0 \le x < \partial^n$

$\downarrow$ $x = x_n \partial^{n-1} + x_2 \partial^{n-2} + \dots + x_n \partial^0$

$$|\psi_3\rangle = (H^{\otimes n} \otimes I) |\psi_2\rangle$$
$$= (H^{\otimes n} \otimes I) \frac{1}{\sqrt{\partial^n}} \sum_x (-1)^{f(x)} |x\rangle |-\rangle$$

using linearity

$$= \frac{1}{\sqrt{\partial^n}} \sum_x (-1)^{f(x)} (H^{\otimes n} |x\rangle) |-\rangle$$

Q: What is $H^{\otimes n}|x\rangle$

$\Rightarrow$ Let us consider an example

$n=1 \quad H|x_1\rangle = \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{x_1 z_1} |z_1\rangle \qquad z_1 \in \{0,1\}$

Two cases:

① $x_1 = 0 \quad H|0\rangle = \frac{1}{\sqrt{2}} \sum_{z_1} |z_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle$

② $x_1 = 1 \quad H|1\rangle = \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{z_1} |z_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$

$n: \quad H^{\otimes n}|x\rangle_n = H^{\otimes n}|x_1\rangle|x_2\rangle \dots |x_n\rangle$

$= (H|x_1\rangle) \otimes (H|x_2\rangle) \otimes \dots \otimes H|x_n\rangle$

$= \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{x_1 z_1} |z_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{z_2} (-1)^{x_2 z_2} |z_2\rangle \otimes \dots \otimes \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{x_n z_n} |z_n\rangle$

$= \frac{1}{\sqrt{2^n}} \sum_{z_1 \dots z_n} (-1)^{x_1 z_1 + x_2 z_2 + \dots + x_n z_n} |z_1, z_2, \dots, z_n\rangle$

$x \cdot z$ (exponent) $\qquad |z\rangle \Rightarrow$ denoted

$\Rightarrow$ two vectors w/ $0 \& 1$

$x = (x_1, x_2, \dots, x_n)$

$z = (z_1, z_2 \dots, z_n)$

$$\frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle$$

Let us plug this into $|\psi_3\rangle$

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} \left(H^{\otimes n}|x\rangle \right) |-\rangle \qquad \text{(replace } H^{\otimes n}|x\rangle \text{)}$$

$$= \sum_z \left(\frac{1}{2^n} \sum_x (-1)^{f(x)} (-1)^{x \cdot z} \right) |z\rangle |-\rangle \Rightarrow \text{expression in computational basis}$$

(order doesn't matter)

$z = 0 \dots 0$

$$P(|0\rangle_n) = \left| \frac{1}{2^n} \sum_x (-1)^{f(x) + x \cdot 0} \right|^2$$

use born's rule

$$P(|0\rangle_n) = \left| \frac{1}{2^n} \sum_x (-1)^{f(x)} \right|^2 \Rightarrow \text{don't have } x \cdot z \text{ term}$$

$f(x)$ constant: $\left| \frac{1}{2^n} \sum_x (-1)^{f(x)} \right|^2$ $\Rightarrow (f(x)=f(0))$

$= \left| \frac{1}{2^n} \sum_x (-1)^{f(x)} \right|^2$

$= \left| \frac{1}{2^n} \sum_x (-1)^{f(0)} \right|^2$ → b/c there is a summation over x

$= \left| \frac{1}{2^n} (-1)^{f(0)} \cdot 2^n \right|^2$

$= \left| (-1)^{f(0)} \right|^2 = 1$

balanced: $x=0$ simplifies the problem :)

$$\left| \frac{1}{2^n} \sum_{x: f(x)=0} (-1)^{f(x)} + \frac{1}{2^n} \sum_{x: f(x)=1} (-1)^{f(x)} \right|^2$$

$= \left| \frac{1}{2^n} \sum_{x: f(x)=0} 1 + \frac{1}{2^n} \sum_{x: f(x)=1} (-1) \right|^2$ (by defn since $f(x)=1$)

$= \left| \frac{1}{2^n} \#\{x \mid f(x)=0\} - \frac{1}{2^n} \#\{x \mid f(x)=1\} \right|^2$ (number of x s.t. $f(x)=0$)

$= 0$

Clicker: How many times do you need to run the algorithm

$\Rightarrow$ Once! → we just take a measurement :)

measure either 1 or 0

Classical Measure $2^{n-1}+1$

$\Rightarrow$ query model / oracle model $\Rightarrow$ Note to Rezy!

- $\Rightarrow$ gives insight
- $\Rightarrow$ black box Review
- only way we can compare these two

$\Rightarrow$ Some algorithms can be equal for classical & quantum but we cannot find an equivalent ex. Shor's Algorithm

$\Rightarrow$ we can find a "quantum mechanism" (idk how else to word it)

New Algorithm!

Bernstein-Vazirani algorithm:

Let $f(x) = a \cdot x = a_1 x_1 \oplus a_2 x_2 \oplus \dots \oplus a_n x_n$

$\Rightarrow$ goal: find description of function $\Rightarrow$ find out what a is

This is an oracle model

We can query the result

$\Rightarrow$ for a given x, it can return $f(x)$ immediately

Def $f(x) = ax = a_1x_1 \oplus a_2x_2 \oplus \dots \oplus a_nx_n$

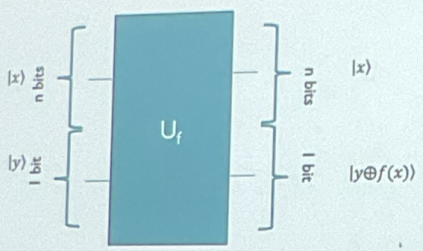
where $a = a_1 a_2 \cdots a_n \Rightarrow$ bitstring $\quad a_i \in \{0,1\}$

$a \in \{0,1\}^n$

f is an oracle / black box

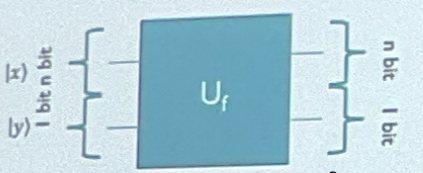
Classical: n queries to determine a)

Unitary computation circuit



$U_f |x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$

The 1st trick



$U_f |x\rangle|-\rangle = (-1)^{f(x)} |x\rangle|-\rangle \Rightarrow$ same trick as Deutsh Joza

$\Rightarrow$ uses interference

Bit-flip $\Rightarrow$ sign change (phase-flip) $\Rightarrow$ to use quantum transformation

Grand trick

$H^{\otimes n} |x\rangle_n$ where $x = x_1, x_2, \dots, x_n$, $z = z_1, z_2 \dots, z_n$

$= \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle_n$ Fourier transformation on $\mathbb{Z}_2^n$ or $\mathbb{F}_2^n$

where $x \cdot z = x_1 z_1 + x_2 z_2 + \dots + x_n z_n = x_1 z_1 \oplus x_2 z_2 \oplus \dots + x_n z_n$ (inner product)

Fourier transformation on $\mathbb{R}$

$(-1)^c = (-1)^{c \bmod 2}$ $c \in \mathbb{N}$ (0 inclusive)

$\hat{f}(k) \propto \int e^{ikx} f(x)\,dx$ (correspondence: k ↔ z, x ↔ x)

⇒ This is a quantum fourier transformation

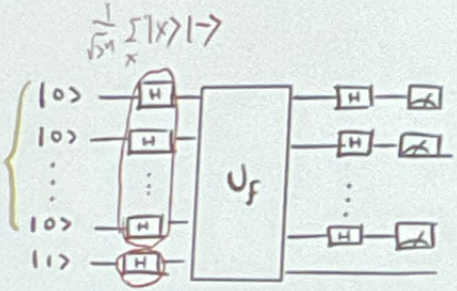
z = position

x = momentum

Quantum fourier transformation

$|k\rangle \propto \int e^{ikx} |x\rangle dx$

Put everything together (the BV algorithm)



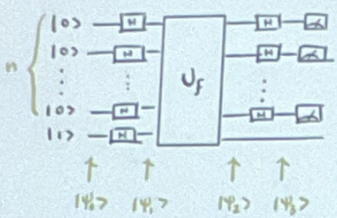
this is a fancy version

analyze $|\psi_2\rangle$ → result after fourier transformation

→ same circuit as Deutsch-Jozsa

$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)}$ → superposition of all inputs

Analysis



$|\psi_3\rangle = (H^{\otimes n} \otimes I) \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle|-\rangle$

$= \frac{1}{\sqrt{2^n}} \sum_x (H^{\otimes n}|x\rangle)\,|-\rangle \cdot (-1)^{f(x)}$

$= \frac{1}{\sqrt{2^n}} \sum_x \frac{1}{\sqrt{2^n}} \sum_z \cdot (-1)^{f(x)} (-1)^{x \cdot z} |z\rangle|-\rangle$

$= \frac{1}{2^n} \sum_z \left(\sum_x (-1)^{f(x)+x \cdot z} \right) |z\rangle|-\rangle$

We want to compute $P(|z\rangle_n) = \left| \frac{1}{2^n} \sum_x (-1)^{f(x)} + xz \right|^2$

$$= \frac{1}{2^n} \left| \sum_x (-1)^{a \cdot x + z \cdot x} \right|^2$$

$$= \frac{1}{2^n} \left| \sum_x (-1)^{(a \oplus z) \cdot x} \right|^2$$

Clicker

$$\sum_{x \in \{0,1\}^n} (-1)^{(a \oplus z) \cdot x} = \prod_{j=0}^{n-1} \sum_{x_j=0}^{1} (-1)^{(a_j \oplus z_j) x_j}$$

If $\exists a_j \neq z_j$, then the equation equals to

(A) 0 ⟹ since there is at least 1 $\emptyset$

B. n

C. something else

if $a_j \neq z_j$

$a_j \oplus z_j = 1$

consider the case where $a_j = z_j$

$$\sum_{x_j=0}^{1} (-1)^{(a_j \oplus z_j) x_j}$$

$$= \sum_{x_j=0}^{1} (-1)^{x_j}$$

$$= (-1)^0 + (-1)^1$$

$$= 1 - 1 = 0$$

Putting everything together

$$P(|z\rangle_n) = \begin{cases} 0 & \text{if } z \neq a \\ 1 & \text{if } z = a \end{cases}$$

since $a \oplus z = 1$

$\therefore \left| \frac{1}{2^n} \sum_x (-1)^0 \cdot \text{stuff} \right|^2 = 1$

Quantum algorithm: 1 → since we need one query

Simons Problem: (Precursor to Shor's Algorithm)

$f: \{0,1\}^n \longrightarrow \{0,1\}^b$

two-to-one function

$f(x_1) = f(x_2) \longleftrightarrow x_1 \oplus b = x_2$ (this is a hidden period) $f(a+T) = f(x)$

GOAL: Find b) (hidden period)

If $y \oplus b \neq x$ $\quad f(x) \neq f(y)$

Clicker

Let be our function as $f:\{0,1\}^n \to \{0,1\}^{n-1}$, $b \neq 0$ before, and the hidden "period". Assume I have asked my oracle to give me f(x) for m different values of x: $x_1, \ldots, x_m$: and I find all queries to give unique results. How many values of b have I ruled out at this stage at most?

$f(x_i) \neq f(x_j)$

$c = x_i \oplus x_j$ ↗ $x_i \oplus c = x_j$ ↘ $x_j \oplus c = x_i$

$\Rightarrow$ c is not period

$\frac{m(m-1)}{2}$ we have $m(m-1)$ pairs (Watch is the $\div 2$)

A. $m(m-1)/2$

B. m^2

C. m

D. 1

Simon's Problem: classically

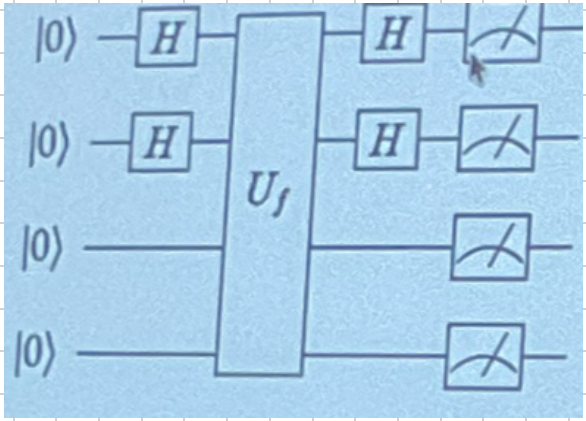
suppose we query m inputs

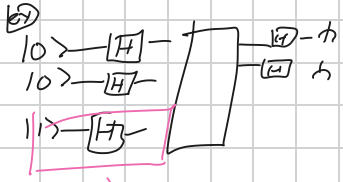
$\frac{m(m-1)}{2} \approx 2^n$ ← at most

Suppose m & n are large s.t $m^2 \approx 2^n$

$m \approx \sqrt{2^{n+1}} = 2^{\frac{n+1}{2}}$

→ classical is exponential time





↳ difference is here for Simon's

$U_f |x\rangle|-\rangle = (-1)^{f(x)} |x\rangle|-\rangle$ → Quantum Parallelism not enough
$U_f |x\rangle|0\rangle = |x\rangle|f(x)\rangle$

The algorithm

I) prepare the state before the quantum oracle

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle_n \otimes |0\rangle_n$$

$$H^{\otimes n} |0\rangle_n$$

II) apply the quantum oracle

$$|\psi_2\rangle = U_f |\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle |f(x)\rangle$$

Measure this

III) measure the second part (output register)

don't need to do it, but it's easier to understand!
(will not cause decoherence other states)